

From Social Reasoning to Embodied Interaction: An Agentic Framework for Social Robots

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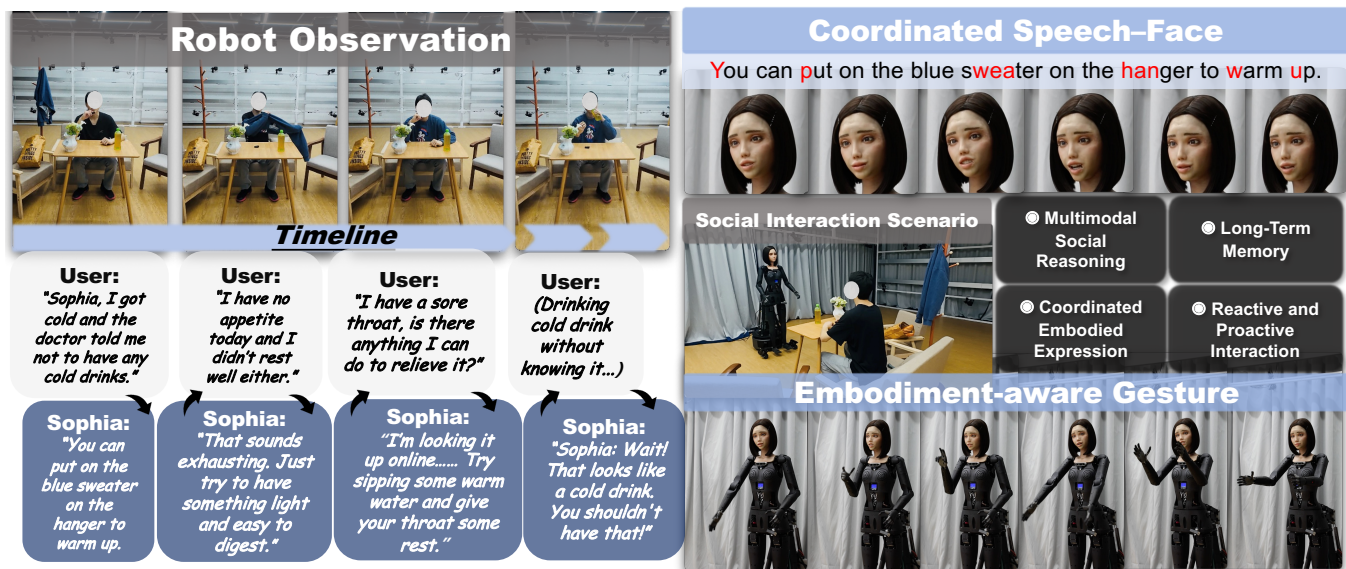


Fig. 1. **Illustration of ARISE.** Given robot observations and multi-round user interaction, Sophia integrates visual understanding, long-term interaction context, proactive engagement, robot-native gesture planning, and coordinated speech-face-gesture execution to produce context-aware embodied responses.

Abstract—Natural face-to-face human-robot interaction requires a robot to understand an evolving social situation, decide when to engage, and express its intent through coordinated physical behavior. Yet existing approaches rarely close this loop: foundation-model agents provide increasingly capable multimodal reasoning and memory but remain largely disembodied, while expressive virtual agents do not face the physical constraints of real robots, and physical social robots typically address social reasoning and embodied expression only partially. We present ARISE, a unified framework that bridges Agentic Reasoning and Interactive Social Embodiment on the Sophia humanoid robot. ARISE integrates multimodal context understanding, long-term memory, and reactive and proactive interaction to determine when and what to communicate, and translates social intent into robot-native gestures coordinated with speech and mechanical facial expressions through streaming execution. Extensive evaluations on Sophia demonstrate strong perceived interaction quality, expressive and well-coordinated embodied behavior, and substantial latency reductions through streaming execution. These results highlight the importance of jointly reasoning about what to communicate, when to engage, and how to physically express social intent for natural interaction with humanoid robots.

I. INTRODUCTION

Empowering embodied agents with social intelligence is central to realizing general-purpose human-robot interaction. A capable social agent should not only infer human intentions from multi-modal interactions and generate appropriate responses, but also continuously perceive and reason about its surroundings, maintain awareness over extended interac-

tions, and proactively engage with humans in real time [9], [5]. These capabilities become particularly compelling when embodied in humanoid robots, where perception, reasoning, communication, and physical behavior are tightly coupled in real-world interaction. Such physically grounded social intelligence therefore represents a promising direction toward interactive embodied agents that can naturally coexist and communicate with humans in everyday environments.

Recent advances in foundation models have substantially expanded the capabilities of digital social agents, enabling natural, context-aware communication and long-horizon multimodal interaction [9]. Yet, such agents remain fundamentally disembodied, lacking the nonverbal behaviors essential to social interaction. Video-based avatars and 3D virtual humans have introduced expressive facial and full-body behaviors, including human animation [10], [11], co-speech gesture generation [1], [3], and real-time motion synthesis [12], [2], [13], but remain confined to virtual spaces. Transferring these capabilities from digital embodiments to physical robots remains challenging: virtual motion is largely unconstrained, whereas physical execution must respect limited degrees of freedom, joint limits, and hardware-specific actuation constraints. Physical social robots have begun to integrate foundation-model reasoning with expressive behavior, supporting affect-aware interaction [14], proactive assistance [5], interruption-aware dialogue [15], and expressive facial behavior [16], [6]. However, these capabilities have

TABLE I
CAPABILITY COMPARISON WITH PREVIOUS WORKS. ✓: DEMONSTRATED; ✗: NOT REPORTED.

Method	Physical robot	Long-term memory	Proactive engagement	Facial expression	Co-speech gestures
EMAGE [1]	✗	✗	✗	✓	✓
Social Agent [2]	✗	✗	✗	✓	✓
MIBURI [3]	✗	✗	✗	✓	✓
Aly and Tapus [4]	✓	✗	✗	✓	✓
Attentive Support [5]	✓	✗	✓	✗	✗
ExFace [6]	✓	✗	✗	✓	✗
Garello et al. [7]	✓	✓	✓	✓	✗
ProAct [8]	✓	✓	✓	✗	✓
Ours	✓	✓	✓	✓	✓

largely been studied in isolation. A unified physical social agent that can understand evolving social context, engage proactively, and coordinate verbal and nonverbal behavior remains largely unexplored.

To address this gap, we present a unified embodied social interaction system deployed on the Sophia [17] humanoid robot. At its core, a multimodal social agent directly reasons over conversational audio and visual observations, augmented with long-term memory, online knowledge retrieval, and proactive interaction to support both reactive and self-initiated engagement. An embodiment-aware gesture planner further translates response intent into verified symbolic gesture programs composed of executable upper-body motion primitives. To realize these high-level decisions as coherent physical behavior, we introduce a streaming multimodal orchestration scheme that synchronizes speech with audio-driven facial animation and schedules gestures online using the audio–facial stream as a shared temporal reference. Together, these components couple multimodal social reasoning with coordinated verbal and nonverbal behavior, enabling context-aware and proactive face-to-face interaction on a physical humanoid robot.

We summarize our core contributions as follows:

- 1) We present *ARISE*, a unified framework for physical humanoid social interaction that combines multimodal social reasoning, long-term memory, and coordinated embodied expression, supporting both reactive and proactive engagement.
- 2) We bridge high-level agentic social reasoning and low-level embodiment-aware gesture planning to translate social intent into executable robot behavior. Streaming Multimodal Orchestration further coordinates asynchronous speech, facial animation, and gesture streams for progressive embodied execution.
- 3) Extensive evaluations on Sophia demonstrate strong context-aware social interaction, effective proactive engagement, and expressive, coordinated embodied behavior.

II. RELATED WORKS

A. Social Reasoning with Foundation Models

Large models have expanded social interaction from dialogue generation toward multimodal understanding, memory, and contextual planning [18]. Socratic Models [19] and embodied multimodal models such as PaLM-E [20] compose pretrained models through language to support multimodal reasoning and assistive dialogue. Generative Agents [21] integrate memory, reflection, and planning to produce sustained social behavior in simulated environments. M3-Agent [9] further extends memory-based reasoning to visual and auditory experience through episodic and semantic memory. Together, these approaches provide increasingly capable foundations for understanding social context and formulating appropriate responses. However, social reasoning alone does not deliver the physical presence and expressive behavior needed for immersive interaction. Our framework bridges this gap by translating multimodal, memory-augmented reasoning into coordinated speech, facial expressions, and upper-body gestures on a physical humanoid.

B. Co-speech Gesture Generation

Co-speech gesture generation has advanced toward expressive, holistic animation of 3D digital humans. EMAGE [1] integrates audio conditioning with masked gesture modeling to generate coordinated face, body, and hand motion. DiffSHEG [22] jointly models facial expressions and body gestures through diffusion, supporting synchronized motion generation for arbitrary-length speech. More recently, MIBURI [3] enables online, causal generation of expressive gestures and facial expressions alongside streaming dialogue. These methods advance motion quality and responsiveness primarily in virtual human representations. However, transferring such expressiveness to physical robots requires accommodating fewer degrees of freedom and hardware-specific joint limits. We address this challenge by composing and verifying robot-native gesture programs to support semantic expressiveness and speech coordination within the robot’s constrained motion repertoire.

C. Multimodal Interaction in Social Robotics

Social robotics has progressed from coordinated multimodal expression toward adaptive, reasoning-driven interaction. Early work by Aly and Tapus [4] combines emotional speech, facial expressions, and speech-accompanying head and arm gestures in narrative interaction. More recent systems advance social decision-making: Attentive Support [5] reasons about when and how to assist human groups, while Garello et al. [7] integrate LLM-based interaction management with knowledge-graph memory for adaptive tutoring. In parallel, speech-driven animatronic control and ExFace improve expressive facial actuation on physical robots [16], [6]. ProAct [8] further connects memory-based proactive reasoning with streaming humanoid motion, bringing social planning and embodied behavior closer together.

However, these systems typically emphasize selected aspects of social reasoning or embodied expression, leaving their integration incomplete, as summarized in Table I. Our framework unifies memory-augmented reactive and proactive planning with coordinated speech, mechanical facial expressions, and verified upper-body gestures through a shared streaming pipeline, enabling context-aware, expressive, and responsive interaction on a physical humanoid.

III. METHOD

A. Overview

As illustrated in Fig. 2, our framework maps ongoing audio–visual interaction to coordinated verbal and nonverbal behaviors on the Sophia [17] human-like robot. Given conversational audio A_t and RGB observation V_t , the *Multimodal Social Agent* generates a verbal response and an executable symbolic gesture plan using multimodal reasoning, long-term memory, online retrieval, and proactive interaction. The *Streaming Multimodal Orchestration* module converts these high-level outputs into synchronized speech, facial animation, and temporally scheduled gestures. The *Robot Deployment* module finally maps the coordinated signals to Sophia’s actuators. This design separates semantic social planning from temporal coordination and low-level execution.

B. Multimodal Social Agent

The *Multimodal Social Agent* operates at the semantic level and produces a verbal response R_t and a symbolic gesture program $\hat{S}_t$. It contains an *Agentic Interaction Planner* for deciding what and when to communicate and an *Embodiment-aware Gesture Planner* for generating and refining executable upper-body behaviors [23], [24], [2].

Agentic Interaction Planner. At interaction step t , the planner directly receives raw conversational audio A_t and the current RGB observation V_t through an audio–visual multimodal foundation model. We denote the current multimodal context by $C_t = (A_t, V_t)$. The planner supports *reactive* interactions triggered by human speech and *proactive* interactions initiated by the robot.

For a reactive interaction, a skill router determines whether long-term memory and online search are needed:

$$D_t = \pi_{\text{skill}}(C_t) = (d_t^{\text{mem}}, d_t^{\text{search}}), \quad (1)$$

where $d_t^{\text{mem}}, d_t^{\text{search}} \in \{0, 1\}$ indicate whether memory and online search are activated, respectively. The memory store M_t preserves salient user information and interaction events across interactions. When enabled, the current context is used to retrieve relevant entries $\tilde{M}_t$. Online search is invoked when information beyond the model’s internal knowledge is required, with retrieved results filtered and summarized into $\tilde{K}_t$. If a skill is not selected, its corresponding context is left empty. The reactive response is then generated as

$$R_t^{\text{react}} = f_{\theta}(C_t, \tilde{M}_t, \tilde{K}_t), \quad (2)$$

where f_{θ} denotes the multimodal response model.

Proactive interaction is evaluated independently of user-initiated dialogue. To reduce unnecessary interruptions, evaluation is enabled only when the human partner is silent, the *Reactive Planner* and execution pipelines are idle, and a minimum cooldown interval has elapsed since the previous dialogue. We denote this eligibility condition by $g_t \in \{0, 1\}$.

Once eligible, a visual social estimator based on a VLM evaluates the current social context from the visual observation. The resulting assessment determines whether proactive interaction is appropriate:

$$z_t^{\text{pro}} = f_{\psi}(V_t; P^{\text{pro}}), \quad z_t^{\text{pro}} \in \{\text{initiate}, \text{wait}\}, \quad (3)$$

where P^{pro} specifies the observable social cues and decision criteria for proactive initiation. After a positive decision, relevant long-term memories $\tilde{M}_t^{\text{pro}}$ are retrieved and supplied only to response generation:

$$R_t^{\text{pro}} = g_{\omega}(V_t, \tilde{M}_t^{\text{pro}}), \quad d_t^{\text{pro}} = 1. \quad (4)$$

Thus, long-term memory affects the content of a proactive response but does not influence whether the robot initiates the interaction.

Embodiment-aware Gesture Planner. The gesture planner converts the social intent of R_t into a symbolic program executable on the robot. We define a semantic keyframe library $K = \{k_i = (n_i, c_i, d_i)\}_{i=1}^N$, where n_i , c_i , and d_i denote the keyframe name, semantic category, and natural-language description, respectively. The description d_i provides a neutral semantic abstraction of the motion, specifying what the keyframe visually conveys and in which social contexts it is generally appropriate, without exposing low-level actuator details to the planner. Each keyframe corresponds to a validated robot posture, while actuator-level mappings are handled during deployment. At a high level, the keyframes are organized into four semantic categories: *Neutral and Recovery*, *Social and Affective*, *Communicative and Presentational*, and *Cognitive and Attentive*.

Rather than generating a single gesture directly, the planner first proposes J candidate programs by reasoning over the response and the semantic keyframe library:

$$\{S_t^{(j)}\}_{j=1}^J = h_{\phi}(R_t, K). \quad (5)$$

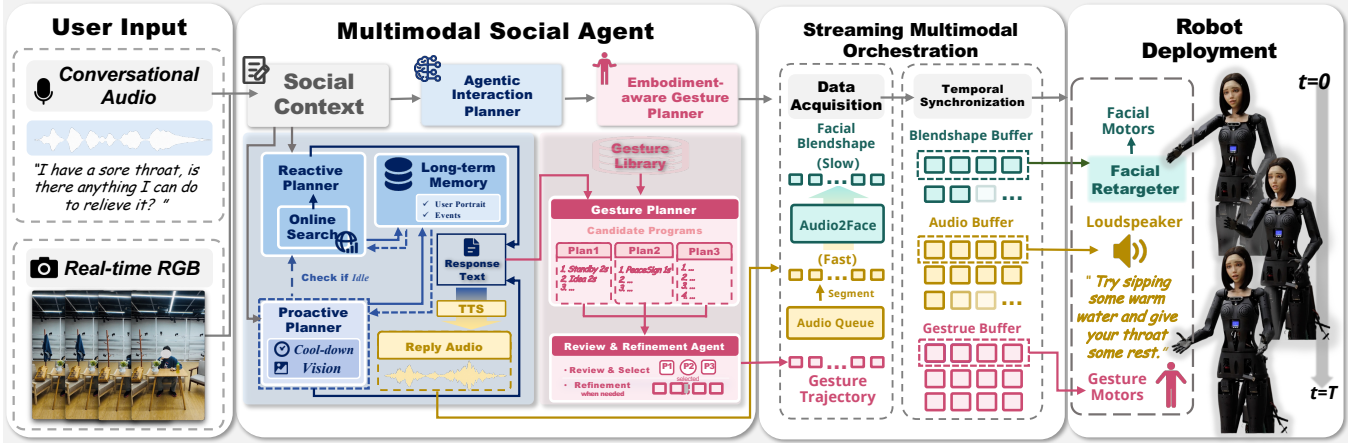


Fig. 2. Overview of ARISE. Given conversational audio and real-time RGB observations, the *Multimodal Social Agent* performs reactive and proactive planning with long-term memory and online search, generating a verbal response and symbolic gesture plan. *Streaming Multimodal Orchestration* synchronizes speech, facial animation, and gestures for progressive execution, and *Robot Deployment* maps the coordinated outputs to robot’s actuators.

Each candidate is a keyframe sequence $S_t^{(j)} = [(n_{j,1}, \tau_{j,1}), \dots, (n_{j,M_j}, \tau_{j,M_j})]$, where $n_{j,m}$ denotes the selected keyframe name and $\tau_{j,m}$ denotes its nominal duration. Candidate construction follows semantic composition rules encoded in the planner, such as pairing hand-shape primitives with a preceding arm raise to preserve visibility, ending most gestures with a neutral standby pose, and avoiding broad presenting motions when inconsistent with the response intent.

The candidate set is then passed to a review-and-refinement agent:

$$\hat{S}_t = g_\eta \left(\{S_t^{(j)}\}_{j=1}^J, R_t, K \right). \quad (6)$$

This stage shares the general motivation of prior work on grounding language-model outputs in executable robot skills, such as SayCan [25], in that unchecked language-model proposals should not be directly executed on physical robots. The agent evaluates the candidates according to semantic fit, social appropriateness, naturalness, timing compatibility, and embodiment feasibility. When a candidate is already suitable, the agent selects it as the final symbolic gesture plan. When minor issues are present, it may refine the sequence, for example by replacing an unsuitable keyframe, adding a neutral ending, or adjusting nominal durations to better match the spoken response. This stage therefore serves both as a candidate-level reviewer and as a lightweight refinement mechanism over the symbolic keyframe space.

The resulting plan $\hat{S}_t = [(n_{t,1}, \tau_{t,1}), \dots, (n_{t,M_t}, \tau_{t,M_t})]$ specifies *what* gesture to perform using validated keyframe names from the semantic library. Low-level actuator commands are not exposed to the planner; they are resolved only during deployment through the predefined mapping associated with each keyframe. The detailed temporal alignment between speech and motion is handled by the subsequent coordination module.

C. Streaming Multimodal Orchestration

This module converts R_t and $\hat{S}_t$ into a coordinated streaming response. It first synchronizes speech and facial animation, then uses the audio–facial stream as the reference clock for gesture scheduling.

The generated speech is streamed as temporally ordered audio chunks $a_{t,r}$. Each chunk is sent in parallel to the audio buffer and Audio2Face, which produces the corresponding facial blendshape chunk

$$f_{t,r} = f_{A2F}(a_{t,r}). \quad (7)$$

Because facial synthesis introduces additional latency, immediate audio playback would cause the facial signal to lag behind speech. We therefore enqueue an audio chunk only after its corresponding facial chunk becomes available. Paired chunks $(a_{t,r}, f_{t,r})$ enter a synchronization buffer, and playback starts after n_{buf} complete pairs are accumulated. The buffer is then drained in temporal order while subsequent pairs are generated and enqueued online. This warm-up strategy trades an initial buffering delay for stable audio–facial synchrony without waiting for the complete response.

The final gesture plan $\hat{S}_t = [(n_{t,1}, \tau_{t,1}), \dots, (n_{t,M_t}, \tau_{t,M_t})]$ is first constrained by a coarse estimate $\hat{T}_t$ of speech duration. Its nominal duration is

$$T_t^g = \sum_{m=1}^M \tau_{t,m}. \quad (8)$$

When T_t^g exceeds the available speech-duration budget, keyframe durations are adjusted before execution to prevent the gesture sequence from substantially exceeding the spoken response. This planning-time adjustment regulates the overall gesture density without imposing fine-grained word-level alignment.

Runtime scheduling is anchored to the actual progress of streaming speech. Let $\delta_{t,r}$ denote the duration of audio chunk $a_{t,r}$. After the r -th chunk, the accumulated audio time is

$$T_{t,r}^{\text{aud}} = \sum_{i=1}^r \delta_{t,i}. \quad (9)$$

Because each facial chunk is paired with the same audio chunk, $T_{t,r}^{\text{aud}}$ defines a shared audio-facial clock. The scheduler maps the ordered keyframe names in $\hat{S}_t$ onto this clock. Each keyframe name $n_{t,m}$ is assigned an execution interval $I_{t,m} = [s_{t,m}, s_{t,m} + \tau_{t,m}]$ using the current audio time and a short look-ahead window to tolerate asynchronous generation and robot playback. Gestures are therefore scheduled progressively as speech is generated rather than after the full response becomes available.

D. Robot Deployment

Deploying ARISE on Sophia requires converting heterogeneous behavioral representations into robot-specific commands while preserving their temporal structure. Speech arrives as audio chunks, facial motion as blendshape sequences, and upper-body behavior as symbolic keyframes, each requiring a different physical realization. Audio is dispatched to the loudspeaker, while Audio2Face outputs are retargeted through ExFace [6] to Sophia’s mechanical facial actuators. Each gesture keyframe is resolved through a pre-defined mapping to a validated upper-body motor target. This representation-specific translation grounds behavior in the robot’s available actuation space without exposing hardware-level control parameters to the language-based planner.

At runtime, the deployment layer dispatches mapped commands according to the shared audio-facial clock established by Streaming Multimodal Orchestration. Audio playback and facial actuation follow the paired chunk sequence, while upper-body motor targets are issued according to their assigned intervals $I_{t,m}$, preserving the planned keyframe order and durations across modality-specific interfaces.

IV. EXPERIMENTS

To validate the performance of the framework and assess its practical applicability to embodied human-robot interaction, we investigate the following research questions:

- Q1.** How effectively does the framework support sustained and context-aware multimodal social interaction?
- Q2.** How effectively does the framework generate expressive robot behaviors that are semantically and temporally aligned with the interaction content?
- Q3.** How effectively do the proposed engineering strategies mitigate latency accumulation across cascaded multimodal processing?

A. Experimental Setup

Hardware Setup. We conduct our experiments on the Sophia [17] humanoid robot developed by Hanson Robotics. An Intel RealSense RGB camera is mounted between Sophia’s head and chest to capture visual observations, and a Philips DLM3545C Bluetooth microphone captures speech. Our framework runs on a host computer equipped with an NVIDIA RTX 3090 GPU, which communicates with Sophia through the robot’s native control interface to deliver speech, facial expressions, and upper-body gestures.

Evaluation Metrics. All ratings were collected using a five-point Likert scale, where 1 denotes “strongly disagree”

and 5 denotes “strongly agree.” We used different rating criteria for the two user studies.

For the *Multi-round Interaction* experiment, participants evaluated each interaction session using four criteria:

- *Overall Interaction Quality* (OIQ): To what extent do the robot’s verbal and non-verbal behaviors together form a coherent and satisfactory social interaction?
- *Contextual Appropriateness* (CA): To what extent is the robot’s response appropriate given the user’s utterance, visible environment, and current interaction context?
- *Conversational Continuity* (CC): To what extent does the robot’s response remain consistent with information and events established earlier in the interaction?
- *Interaction Engagement* (IE): To what extent did the interaction maintain your interest and attention?

For the *Gesture Generation* experiment, participants evaluated each interaction session using four criteria:

- *Motion Naturalness*: To what extent do the robot’s movements appear smooth, physically plausible, and natural?
- *Motion Expressiveness*: To what extent do the robot’s movements convey the intended social or communicative meaning?
- *Speech-Motion Coordination*: To what extent are the robot’s movements temporally and semantically coordinated with its speech?
- *Overall Quality*: Considering both the robot’s speech and movements, how would you rate the overall quality of this response?

User Study. We recruited 20 participants who had no prior knowledge of our system for each of Q1 and Q2. Each participant completed the full evaluation for all configurations within the corresponding experiment.

For Q1, participants were provided with a predefined dialogue script to ensure comparable interaction content, while retaining the freedom to ask natural follow-up questions based on the robot’s responses. For Q2, participants were provided with predefined dialogue scripts and were asked to follow the scripts while focusing on the robot’s verbal and gestural responses.

After each interaction, participants rated the robot according to the evaluation metrics associated with the corresponding study. The order of configurations was independently randomized for each participant. Participants were not allowed to see or interact with the robot before the experiment, and not informed of the configuration used in each interaction or whether it corresponded to the baseline or an ablation. Before each interaction, the robot was reset to the same predefined initial state.

Statistical Analysis. For each configuration, mean and standard deviation were computed across the 20 participant ratings. We applied two-sided paired Wilcoxon signed-rank tests [26] to participant-level scores. P -values were adjusted using the Holm procedure [27]. Significance was denoted by * for $P_{\text{adj}} < 0.05$, ** for $P_{\text{adj}} < 0.01$, and *** for $P_{\text{adj}} < 0.001$.

TABLE II
RESULTS FOR MULTI-ROUND SOCIAL INTERACTION.

Method	OIQ(↑)	CA(↑)	CC(↑)	IE(↑)
Full System	4.38 ± 0.41	4.34 ± 0.43	4.25 ± 0.46	4.31 ± 0.44
w/o Visual Input	3.72 ± 0.58***	2.90 ± 0.69***	4.08 ± 0.50	4.06 ± 0.52
w/o Long-Term Memory	3.54 ± 0.61***	3.78 ± 0.57**	3.29 ± 0.68***	3.62 ± 0.64**
w/o Proactive Interaction	3.76 ± 0.51**	4.23 ± 0.44	4.16 ± 0.47	3.12 ± 0.71***

B. Baselines and Ablation

To assess the contributions of individual components to interaction-response and embodied-gesture quality, we evaluated the Full System together with the following controlled ablation settings.

For Q1, we compare four configurations: **Full System**, which enables visual input, long-term memory, and proactive interaction; **w/o Visual Input**, which removes the current RGB observation; **w/o Long-Term Memory**, which disables memory retrieval and update; and **w/o Proactive Interaction**, which disables proactive behaviors while retaining user-initiated response generation.

For Q2, we compare four configurations of our *Embodiment-Aware Gesture Generation* framework. **Full System** denotes the complete configuration. **Single Candidate** changes the number of generated candidates from $J = 3$ to $J = 1$. **w/o Review-and-Refinement Agent** removes the review-and-refinement stage and directly generates and executes a single gesture sequence. We also include EMAGE [1] as an external speech-driven gesture baseline. EMAGE generates human-body motion in the SMPL-X representation. We retain only its upper-body motion and project it into Sophia’s joint space. Since Sophia’s joints are modeled as single-degree-of-freedom actuators, this projection reduces the corresponding SMPL-X motion to one-dimensional joint commands. Before execution, a safety check clips any command that exceeds the allowable joint range of the robot.

C. Implementation Details and Results

Implementation Details. In the experiment, the *Reactive Planner*, the maintenance *Long-term Memory*, the visual social estimator and the memory retrieved response generation of *Proactive Planner* in our framework are implemented using Doubao-Seed-2.1-Pro-260628. The Audio2Face module employs the NVIDIA Audio2Face-3D v2.3 model [28], and the facial retargeter employs the ExFace [6] model. For gesture generation, the language-agent modules are implemented using DeepSeek-V4. The final gesture sequence is compiled into robot actuator commands during deployment. In the Full System, the planner generates $J = 3$ candidate symbolic gesture programs for each response. The full executable library contains 37 keyframes and consists of 2 *Neutral and Recovery* keyframes, 9 *Social and Affective* keyframes, 24 *Communicative and Presentational* keyframes, and 2 *Cognitive and Attentive* keyframes. Each keyframe is associated with a validated robot posture and a semantic description used by the gesture planner. The review-and-refinement agent scores these candidates, selects the best one, and refines it when necessary.

TABLE III
RESULTS FOR THE GESTURE-GENERATION EXPERIMENT.

Method	Motion Natural.	Motion Express.	Speech-Motion Coord.	Overall Quality
Full System	3.92 ± 0.34	4.22 ± 0.34	4.03 ± 0.60	4.09 ± 0.26
Single Candidate	3.63 ± 0.41**	3.88 ± 0.43*	3.79 ± 0.41	3.92 ± 0.29
w/o R-R Agent	3.00 ± 0.39***	3.35 ± 0.46***	3.21 ± 0.44**	3.58 ± 0.24***
EMAGE	3.37 ± 0.27***	3.67 ± 0.38***	3.89 ± 0.33	3.98 ± 0.25

TABLE IV
LATENCY DEFINITIONS.

Latencies	Definition
L_{MSA}	The latency of <i>Multimodal Social Agent</i> .
L_{ES}	The latency of embodied synthesis.
L_{robot}	The latency of <i>Robot Deployment</i> .
L_{AF}	The end-to-end latency from the end of user’s speech to the beginning of audio-facial playback.

Multi-round Interaction (Q1) Fig. 3 presents some characteristic interaction sequences, including online searching, memory-supported conversational continuity and visually grounded proactive assistance. As shown in Table II, the Full System achieves the best overall performance across all criteria. Removing visual input causes CA to decrease sharply, showing that visual information is essential for understanding interaction situation. Removing long-term memory lowers all four scores. This shows that interaction understanding, especially proactive behavior, depends substantially on contextual information. Underlying language model has some basic memory capability, but cannot maintain the persistent and precise contextual connections provided by the long-term memory module. Without proactive interaction, IE and OIQ decrease, while CA and CC remain close to the Full System. This shows that proactive functions, such as reminders and corrections, primarily improve user engagement and the practical value of the interaction.

Gesture Generation (Q2). Figure 4 shows representative social-aware interaction examples. As shown in Table III, multi-candidate generation and review-and-refinement provide complementary benefits for embodied gesture generation. The performance gap between the Full System and Single Candidate indicates that generating multiple candidates gives the reviewer a broader range of gesture programs to evaluate, leading to more suitable embodied responses. Meanwhile, the performance of Single Candidate relative to w/o Review-and-Refinement Agent shows that reviewing and refining the generated program remains beneficial even when only one candidate is available. EMAGE achieves competitive performance, but its lower motion naturalness and motion expressiveness suggest that embodiment-aware planning with executable gesture library is better suited to the physical robot.

D. More Experiments

System Latency Evaluation (Q3) For Q3, we compare the latency of our proposed *Streaming Multimodal Orchestration* with the *Serial Embodiment Processing* baseline, which generates reply audio and facial animations serially. The benchmark uses 40 short dialogue segments sampled from



Fig. 3. **Multi-round Interaction Sequences** (a) **Birthday Gift**: Sophia preserves confidential information for individual users and remembers the gift’s specific location. (b) **Whose Bag**: Leveraging visual perception to distinguish different users and backpacks, Sophia recalls object-user associations and answers identity-related queries. (c) **Clean up Desk**: Sophia can offer appropriate contextual guidance for object placement, and give positive feedback proactively after the user finishes the task.

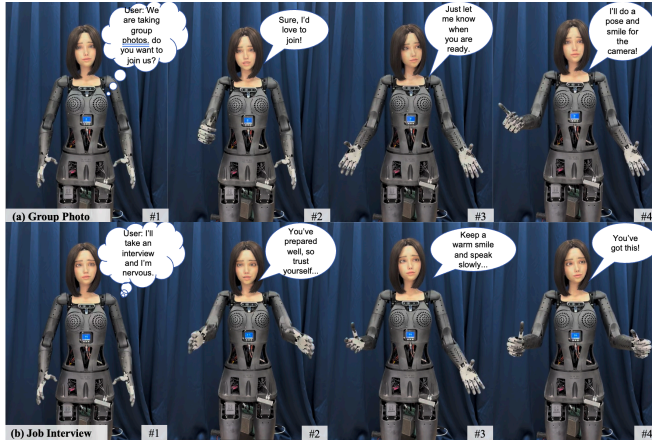


Fig. 4. **Gesture Generation Sequences** (a) **Group Photo**: The robot actively responds to social invitations and cooperates for group-photo scenarios. (b) **Job Interview**: Detecting user nervousness, Sophia delivers comforting verbal feedback and appropriate expressive gestures.

TABLE V

RESULTS FOR SYSTEM LATENCY EVALUATION. VALUES ARE REPORTED AS MEAN.

Method	L_{MSA} (↓)	L_{ES} (↓)	L_{robot} (↓)	L_{AF} (↓)
Streaming Multimodal Orchestration (Ours)	3.306	1.906	0.003	5.214
Serial Embodiment Processing	3.227	10.621	0.003	13.851

the participant interactions. Table IV summarizes the latency definitions. As shown in Table V, the latency of *Serial Embodiment Processing* is far beyond the responsiveness required for natural HRI. In contrast, our proposed *Streaming Multimodal Orchestration* substantially reduces the system delay.

Effect of Executable Gesture Library Size. We further evaluate the effect of gesture library size using five keyframe libraries containing 8, 15, 30, 37, and 45 keyframes, respectively. The following category counts are reported in the order of *Neutral and Recovery*, *Social and Affective*, *Communicative and Presentational*, and *Cognitive and Attentive*.

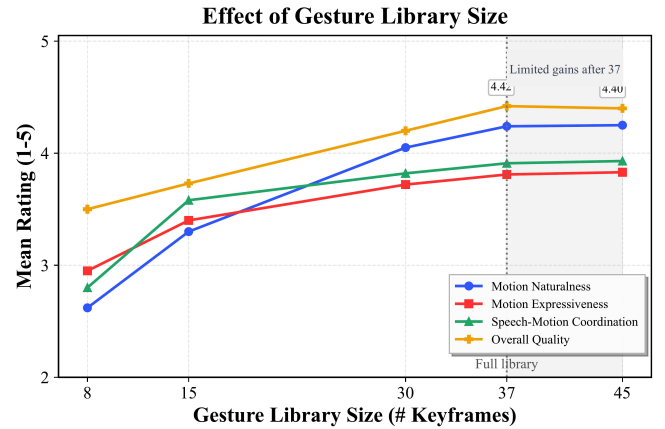


Fig. 5. Effect of gesture library size on user-rated motion quality. Library size is measured by the number of semantic keyframes. The scores increase substantially as the library grows from 8 to 37 keyframes, while the gain becomes marginal when further expanded to 45 keyframes.

The 8-keyframe library contains (1,4,2,1) keyframes across these four categories, respectively. The 15-keyframe library expands these counts to (2,5,7,1), while the 30-keyframe library further expands them to (2,8,18,2). The 37-keyframe library used in the main experiments contains (2,9,24,2) keyframes. Finally, the 45-keyframe library further adds *Social and Affective* and *Communicative and Presentational* motions, resulting in (2,12,29,2) keyframes across the four categories. All other components, including gesture planning, review-and-refinement, and streaming orchestration, remain unchanged.

As shown in Fig. 5, increasing the keyframe library from 8 to 15, 30, and 37 keyframes leads to clear gains in motion naturalness, motion expressiveness, speech-motion coordination, and overall quality. Beyond 37 keyframes, the effect of further expanding the library becomes substantially weaker, showing only marginal changes and no consistent additional improvement.

V. LIMITATIONS

The current cascaded architecture still limits sub-second conversational responsiveness, with most latency arising from multimodal-agent inference and streaming facial synthesis rather than robot execution. Fine-grained facial expressiveness also remains limited on the current hardware platform. Mechanical facial actuation constrains the subtle motions that can be physically realized, introducing platform-dependent limits distinct from the proposed social planning and orchestration mechanisms. Future work will explore lightweight models specialized for social interaction and robot-native facial synthesis to further reduce response latency and improve embodied responsiveness.

VI. CONCLUSION

We presented ARISE, a unified framework that connects multimodal social reasoning with coordinated embodied interaction on the Sophia humanoid. By integrating memory-augmented reactive and proactive interaction, robot-native gesture planning, and streaming speech-face-gesture orchestration, ARISE improves perceived interaction and gesture quality while reducing latency relative to serial embodiment processing. These results highlight the value of jointly considering what and when to communicate and how to express that intent through the robot’s physical capabilities.

REFERENCES

- [1] H. Liu, Z. Zhu, G. Becherini, Y. Peng, M. Su, Y. Zhou, X. Zhe, N. Iwamoto, B. Zheng, and M. J. Black, “Emage: Towards unified holistic co-speech gesture generation via expressive masked audio gesture modeling,” in *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2024, pp. 1144–1154.
- [2] Z. Zhang, Y. Zhou, H. Yao, T. Ao, X. Zhan, and L. Liu, “Social agent: Mastering dyadic nonverbal behavior generation via conversational llm agents,” in *Proceedings of the SIGGRAPH Asia 2025 Conference Papers*, 2025, pp. 1–12.
- [3] M. H. Mughal, R. Dabral, V. Demberg, and C. Theobalt, “Miburi: Towards expressive interactive gesture synthesis,” *arXiv preprint arXiv:2603.03282*, 2026.
- [4] A. Aly and A. Tapus, “Multimodal adapted robot behavior synthesis within a narrative human-robot interaction,” in *2015 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2015, pp. 2986–2993.
- [5] D. Tanneberg, F. Ocker, S. Hasler, J. Deigmoeller, A. Belardinelli, C. Wang, H. Wersing, B. Sendhoff, and M. Gienger, “To help or not to help: Llm-based attentive support for human-robot group interactions,” in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2024, pp. 9130–9137.
- [6] D. Zhang, J. Peng, Y. Jiao, J. Gu, J. Yu, and J. Chen, “Exface: Expressive facial control for humanoid robots with diffusion transformers and bootstrap training,” in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2025, pp. 20 263–20 268.
- [7] L. Garello, G. Belgiovine, G. Russo, F. Rea, and A. Sciutti, “Building knowledge from interactions: An llm-based architecture for adaptive tutoring and social reasoning,” in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2025, pp. 9239–9246.
- [8] Z. Zhang, Z. Kang, R. Zhao, Y. Feng, B. Jiang, and L. Liu, “Proact: A dual-system framework for proactive embodied social agents,” *arXiv preprint arXiv:2602.14048*, 2026.
- [9] L. Long, Y. He, W. Ye, Y. Pan, Y. Lin, H. Li, J. Zhao, and W. Li, “Seeing, listening, remembering, and reasoning: A multimodal agent with long-term memory,” in *International Conference on Learning Representations*, vol. 2026, 2026, pp. 146 197–146 246.
- [10] G. Lin, J. Jiang, J. Yang, Z. Zheng, C. Liang, Y. Zhang, and J. Liu, “OmniHuman-1: Rethinking the scaling-up of one-stage conditioned human animation models,” in *2025 IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE, 2025, pp. 13 848–13 858.
- [11] Z. Sun, Z. Peng, Y. Ma, Y. Chen, Z. Zhou, Z. Zhou, G. Zhang, Y. Zhang, Y. Zhou, Q. Lu *et al.*, “StreamAvatar: Streaming diffusion models for real-time interactive human avatars,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026, pp. 10 887–10 897.
- [12] K. Ji, Y. Shi, Z. Jin, K. Chen, L. Xu, Y. Ma, J. Yu, and J. Wang, “Towards immersive human-x interaction: A real-time framework for physically plausible motion synthesis,” in *2025 IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE, 2025, pp. 10 173–10 183.
- [13] W. Jiang, Y. Xie, H. Fan, Y. Chen, X. Wang, Y. Shi, and J. Wang, “Super star: Towards streaming real-time interactive agents for digital humans,” *arXiv preprint arXiv:2608.24909*, 2026.
- [14] P. Li, L. Cao, X.-M. Wu, X. Yu, and R. Yang, “Ugotme: An embodied system for affective human-robot interaction,” in *2025 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2025, pp. 5542–5548.
- [15] S. Cao, J. Moon, A. Mahmood, V. N. Antony, Z. Xiao, A. Liu, and C.-M. Huang, “Interruption handling for conversational robots,” *arXiv preprint arXiv:2501.01568*, 2025.
- [16] B. Li, H. Li, and H. Liu, “Driving animatronic robot facial expression from speech,” in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2024, pp. 7012–7019.
- [17] D. Hanson, A. Imran, A. Vellanki, and S. Kanagaraj, “A neuro-symbolic humanlike arm controller for sophia the robot,” *arXiv preprint arXiv:2010.13983*, 2020.
- [18] C. Zhang, J. Chen, J. Li, Y. Peng, and Z. Mao, “Large language models for human-robot interaction: A review,” *Biomimetic Intelligence and Robotics*, vol. 3, no. 4, p. 100131, 2023.
- [19] A. Zeng, M. Attarian, B. Ichter, K. Choromanski, A. Wong, S. Welker, F. Tombari, A. Purohit, M. Ryoo, V. Sindhwani *et al.*, “Socratic models: Composing zero-shot multimodal reasoning with language,” *arXiv preprint arXiv:2204.00598*, 2022.
- [20] D. Driess, F. Xia, M. S. Sajjadi, C. Lynch, A. Chowdhery, B. Ichter, A. Wahid, J. Tompson, Q. Vuong, T. Yu *et al.*, “Palm-e: An embodied multimodal language model,” *arXiv preprint arXiv:2303.03378*, 2023.
- [21] J. S. Park, J. O’Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, “Generative agents: Interactive simulacra of human behavior,” in *Proceedings of the 36th annual acm symposium on user interface software and technology*, 2023, pp. 1–22.
- [22] J. Chen, Y. Liu, J. Wang, A. Zeng, Y. Li, and Q. Chen, “Diffshg: A diffusion-based approach for real-time speech-driven holistic 3d expression and gesture generation,” in *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2024, pp. 7352–7361.
- [23] Y. Yoon, W.-R. Ko, M. Jang, J. Lee, J. Kim, and G. Lee, “Robots learn social skills: End-to-end learning of co-speech gesture generation for humanoid robots,” in *2019 International conference on robotics and automation (ICRA)*. IEEE, 2019, pp. 4303–4309.
- [24] C.-M. Huang and B. Mutlu, “Robot behavior toolkit: generating effective social behaviors for robots,” in *Proceedings of the seventh annual ACM/IEEE international conference on Human-Robot Interaction*, 2012, pp. 25–32.
- [25] M. Ahn, A. Brohan, N. Brown, Y. Chebotar, O. Cortes, B. David, C. Finn, C. Fu, K. Gopalakrishnan, K. Hausman *et al.*, “Do as i can, not as i say: Grounding language in robotic affordances,” *arXiv preprint arXiv:2204.01691*, 2022.
- [26] F. Wilcoxon, “Individual comparisons by ranking methods,” *Biometrics bulletin*, vol. 1, no. 6, pp. 80–83, 1945.
- [27] S. Holm, “A simple sequentially rejective multiple test procedure,” *Scandinavian journal of statistics*, pp. 65–70, 1979.
- [28] C. Chung, I. Fedorov, M. Huang, A. Karmanov, D. Korobchenko, R. Ribera, Y. Seol *et al.*, “Audio2face-3d: Audio-driven realistic facial animation for digital avatars,” *arXiv preprint arXiv:2508.16401*, 2025.